

Lab 3

Part 1

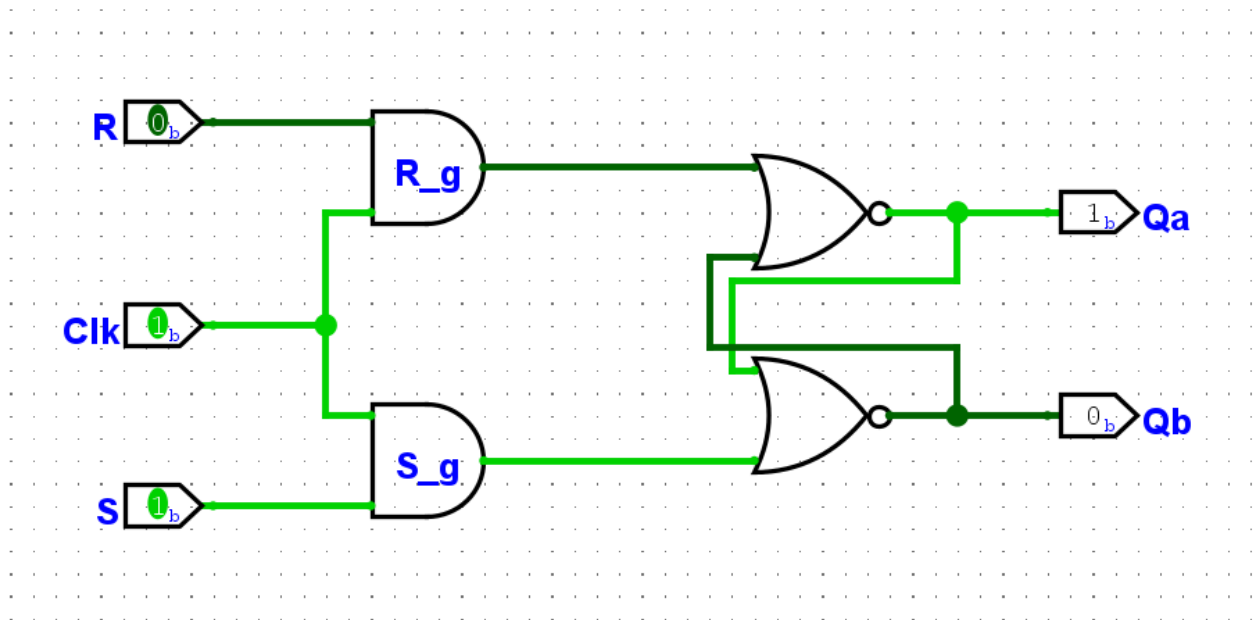


Figure 1 - Logisim RS Latch

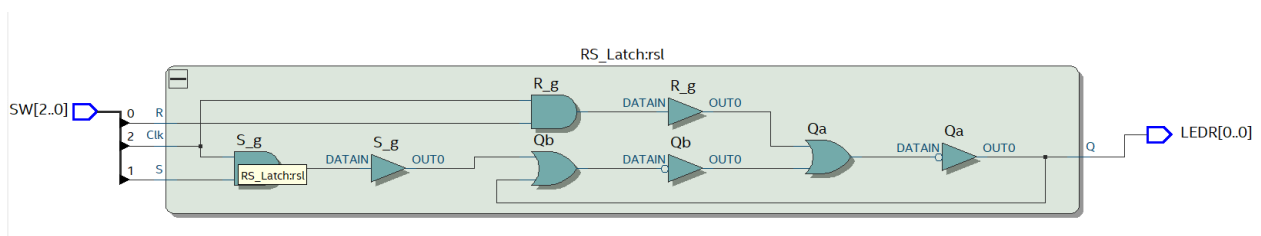


Figure 2 - Quartus RS Latch RTL view

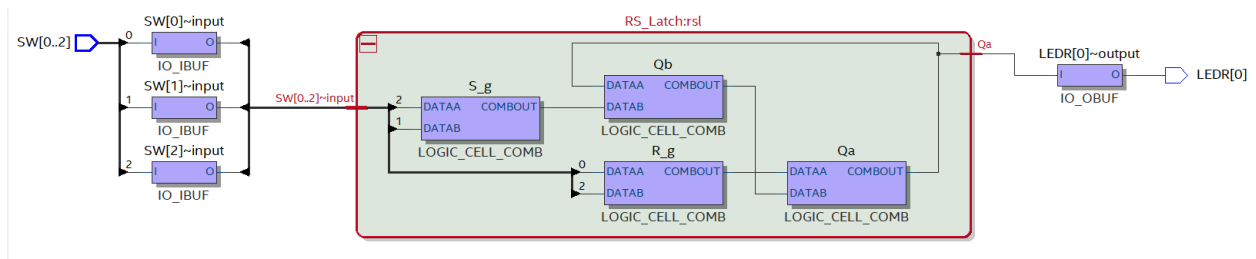


Figure 3 - Quartus RS Latch Technology view



Figure 4 - RS Latch successfully storing high signal

Part 2

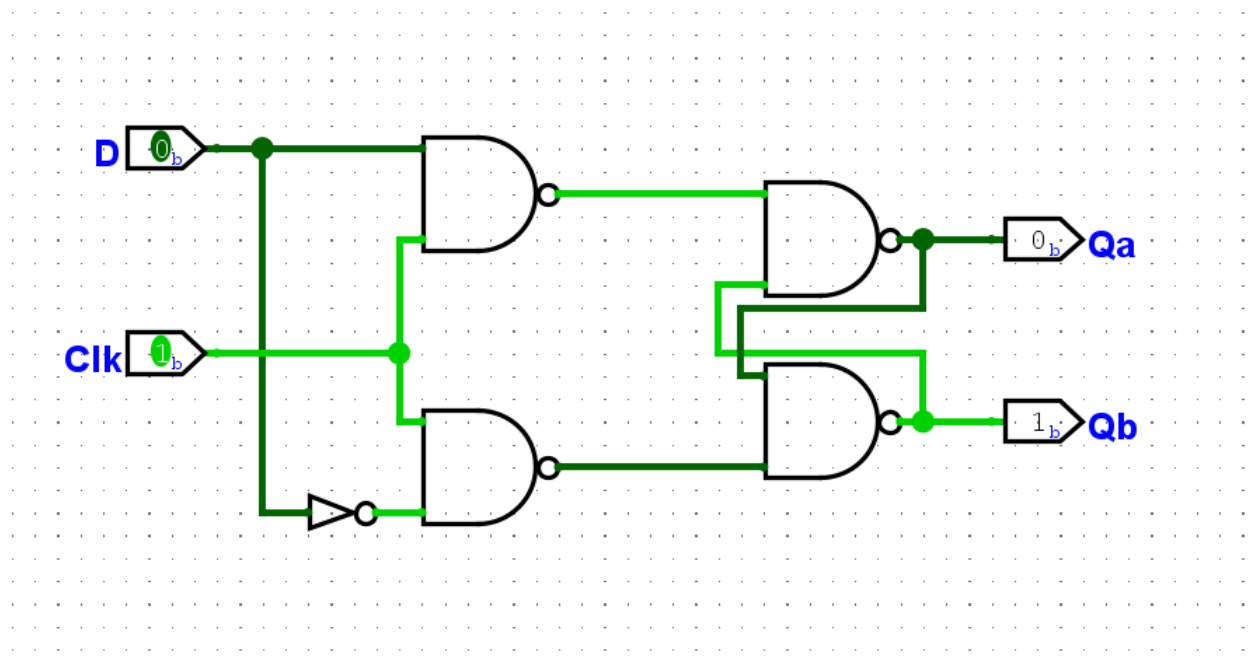


Figure 5 - Logisim Gated D Latch

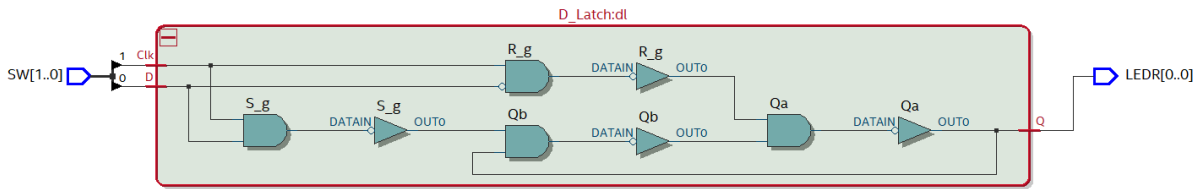


Figure 6 - Quartus Gated D Latch RTL view

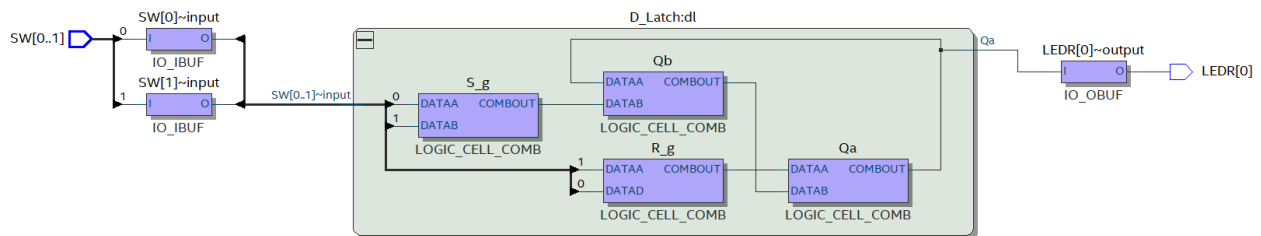


Figure 7 - Quartus Gated D Latch Technology view

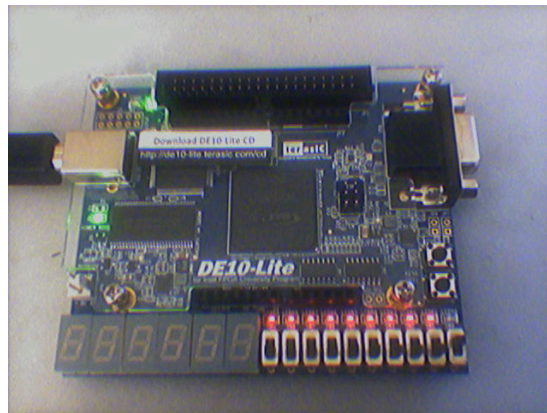


Figure 8 - Clock signal (SW1) is low, thus Gated D Latch does not allow high signal

Part 3

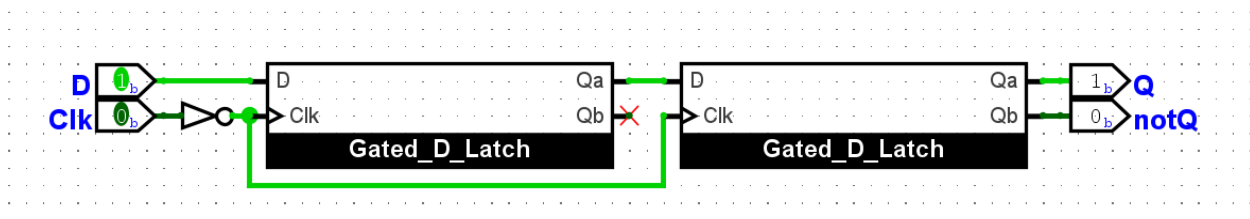


Figure 9 - Logisim Master-Slave D Flop-Flop

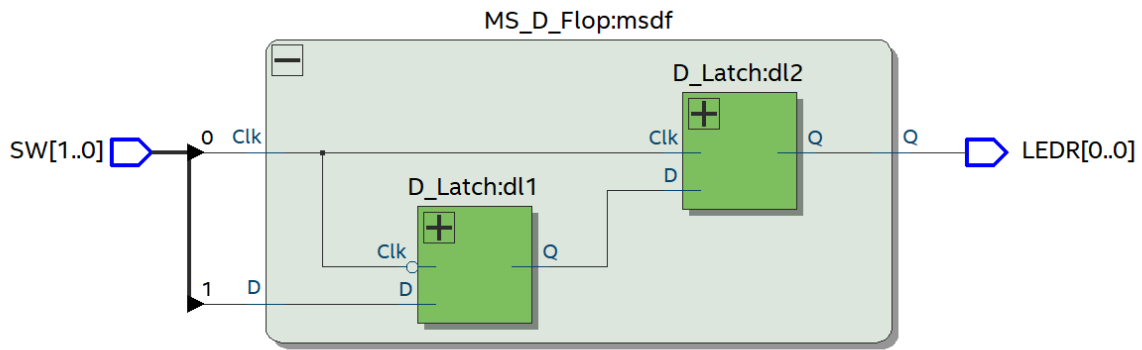


Figure 10 - Quartus Master-Slave D Flip-Flop RTL view

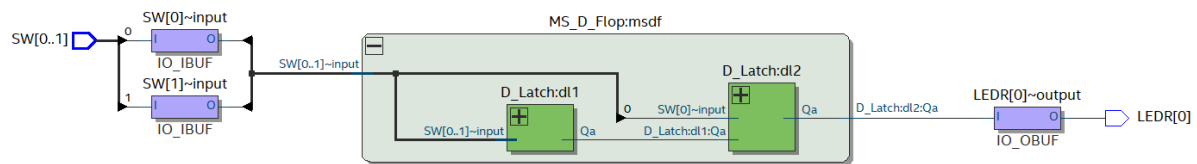


Figure 11 - Quartus Master-Slave D Flip-Flop Technology view

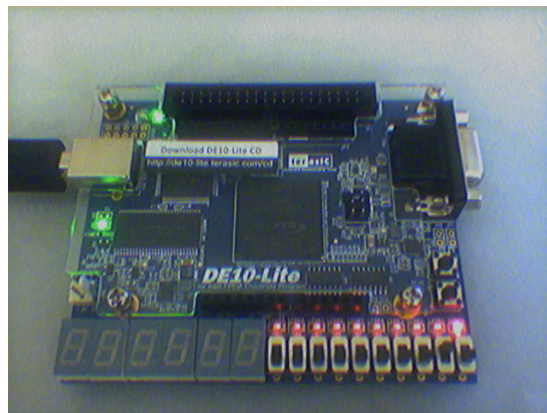


Figure 12 - Master-Slave D Flip-Flop changes output on signal edge compared to RS and D Latch

Part 4

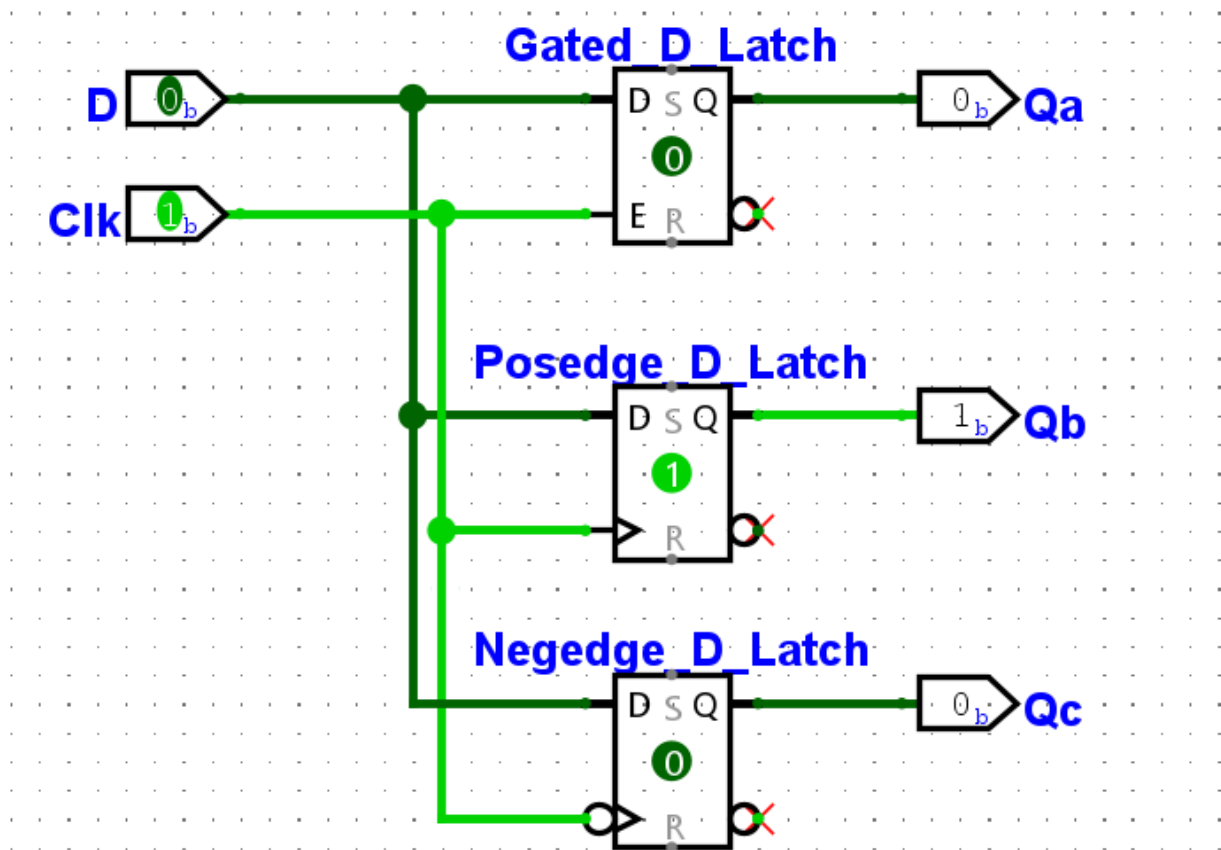


Figure 13 - Logisim active high, positive edge, and negative edge triggered latch

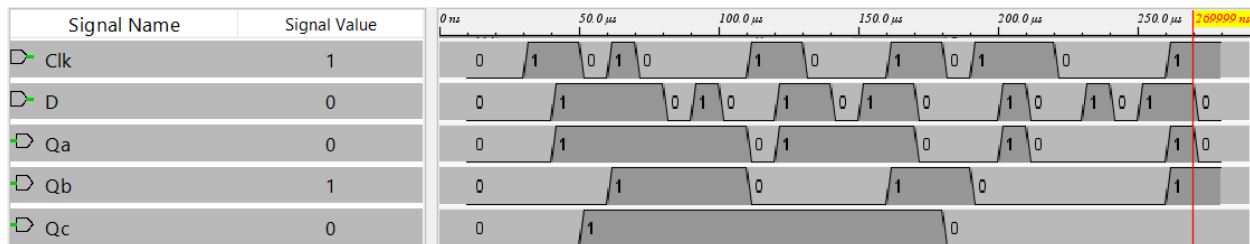


Figure 14 - Could not get Qsim to work. Logisim timing diagram shows correct behavior.

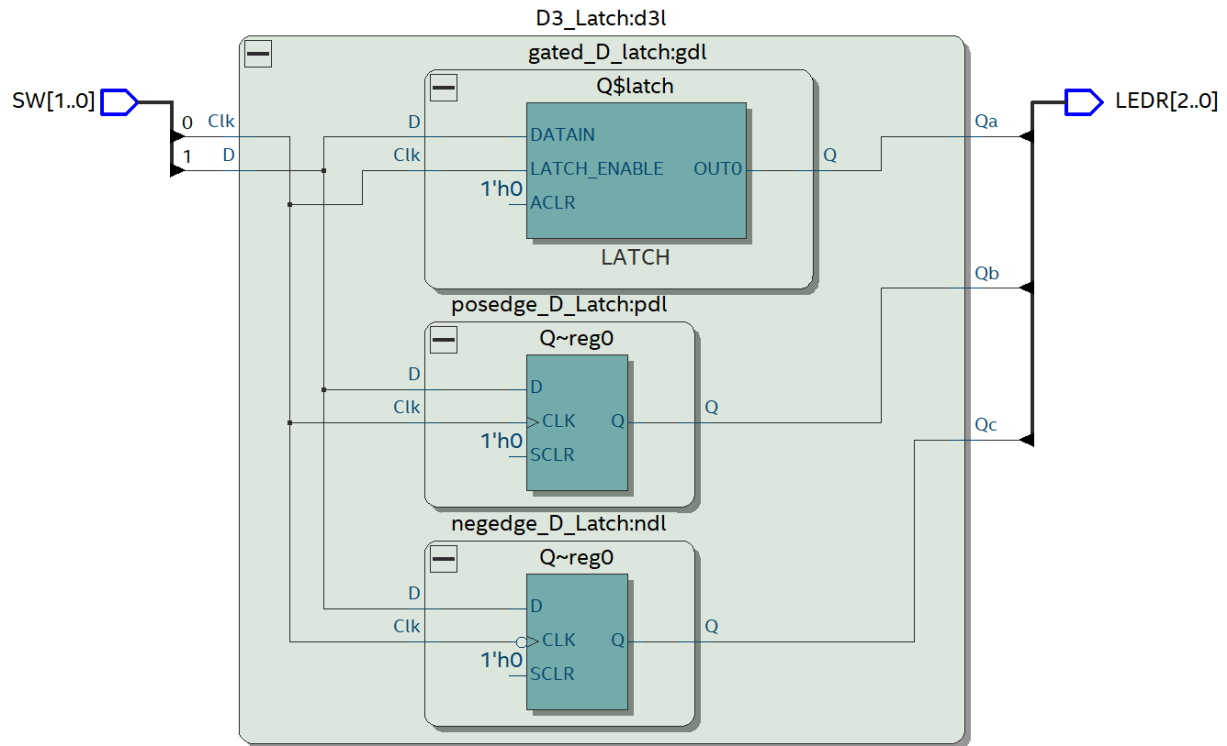


Figure 15 - Three Latch Quartus RTL view

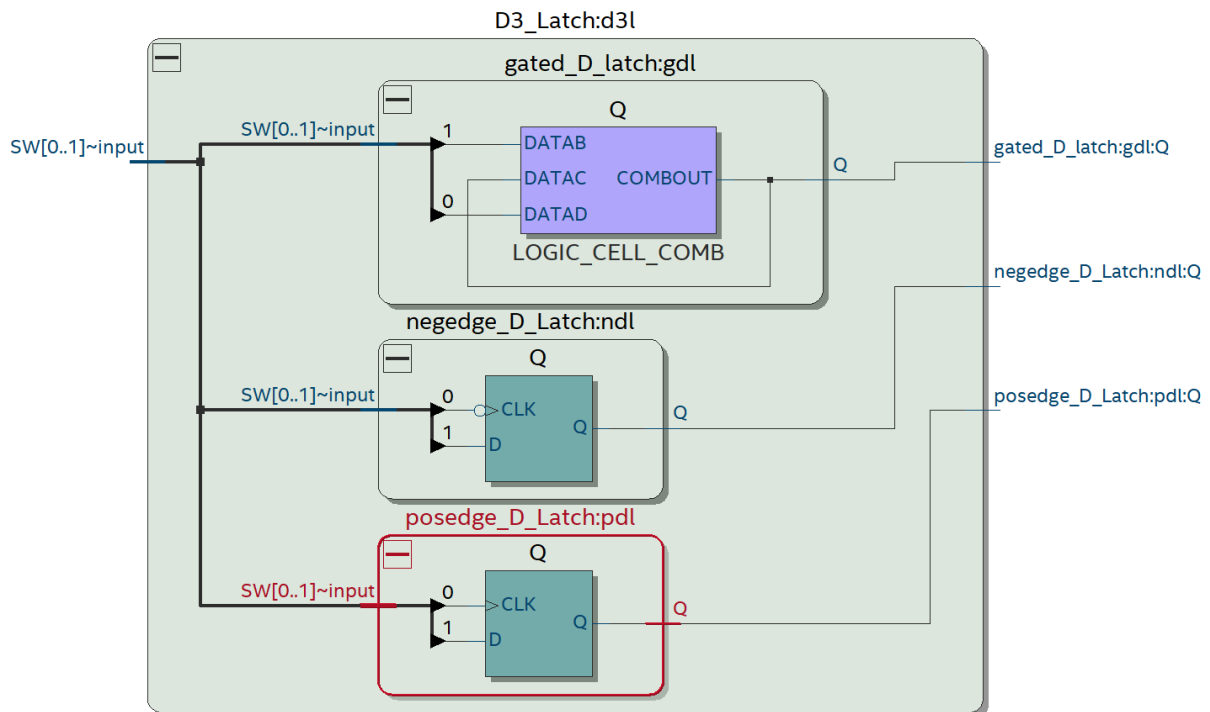


Figure 16 - Three Latch Quartus Technology view

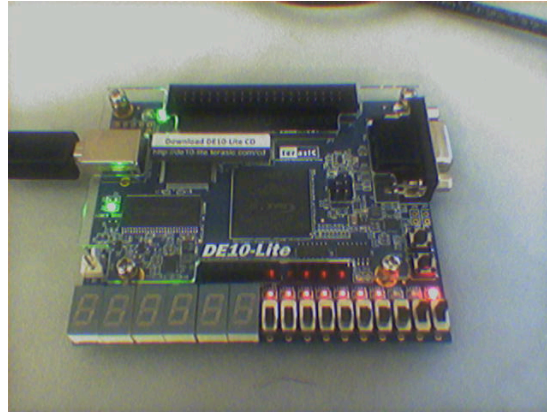


Figure 17 - Only Active High LED glows when Clk is high and D goes high.

Part 5

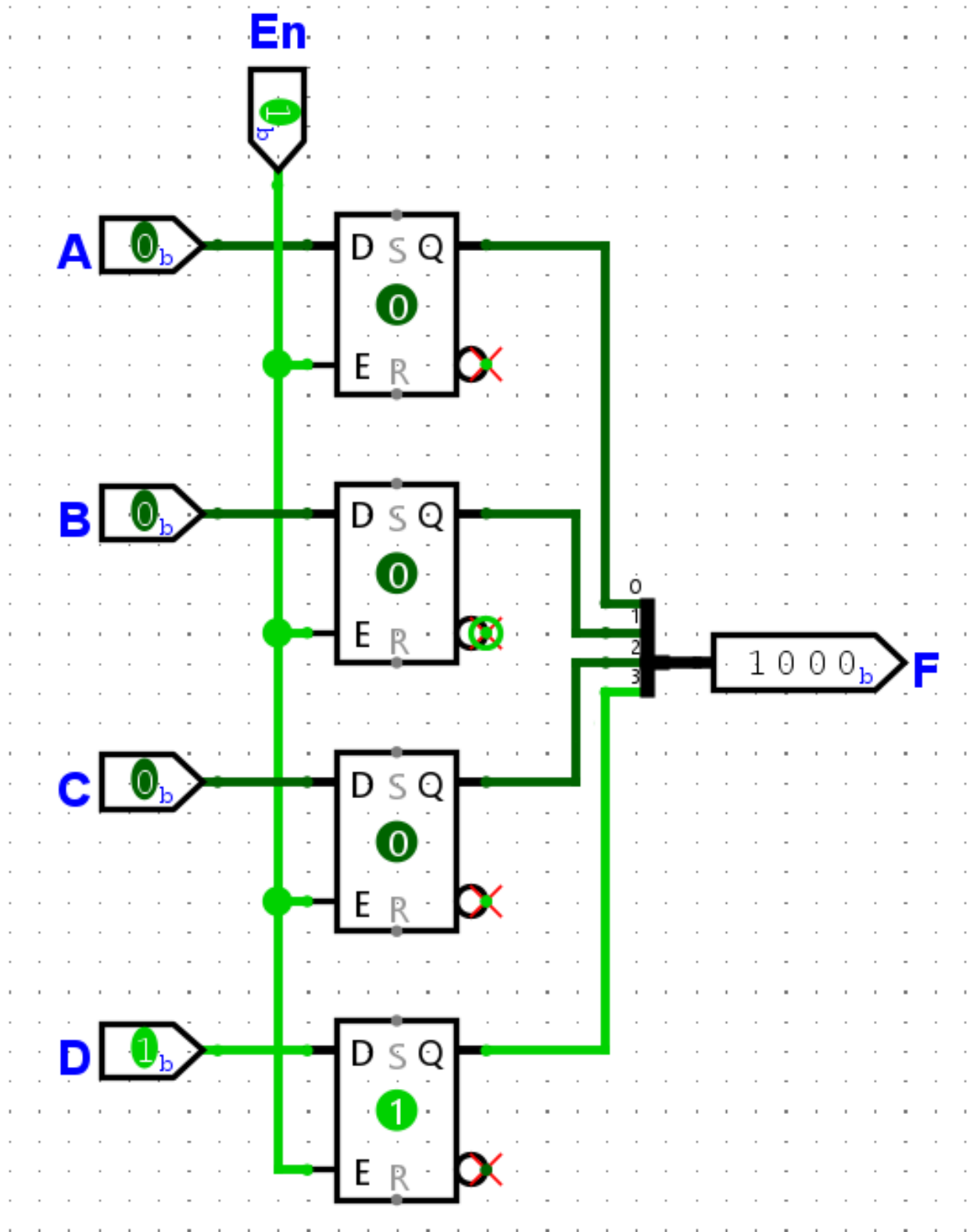


Figure 18 - Logisim 4-Bit storage active high latch

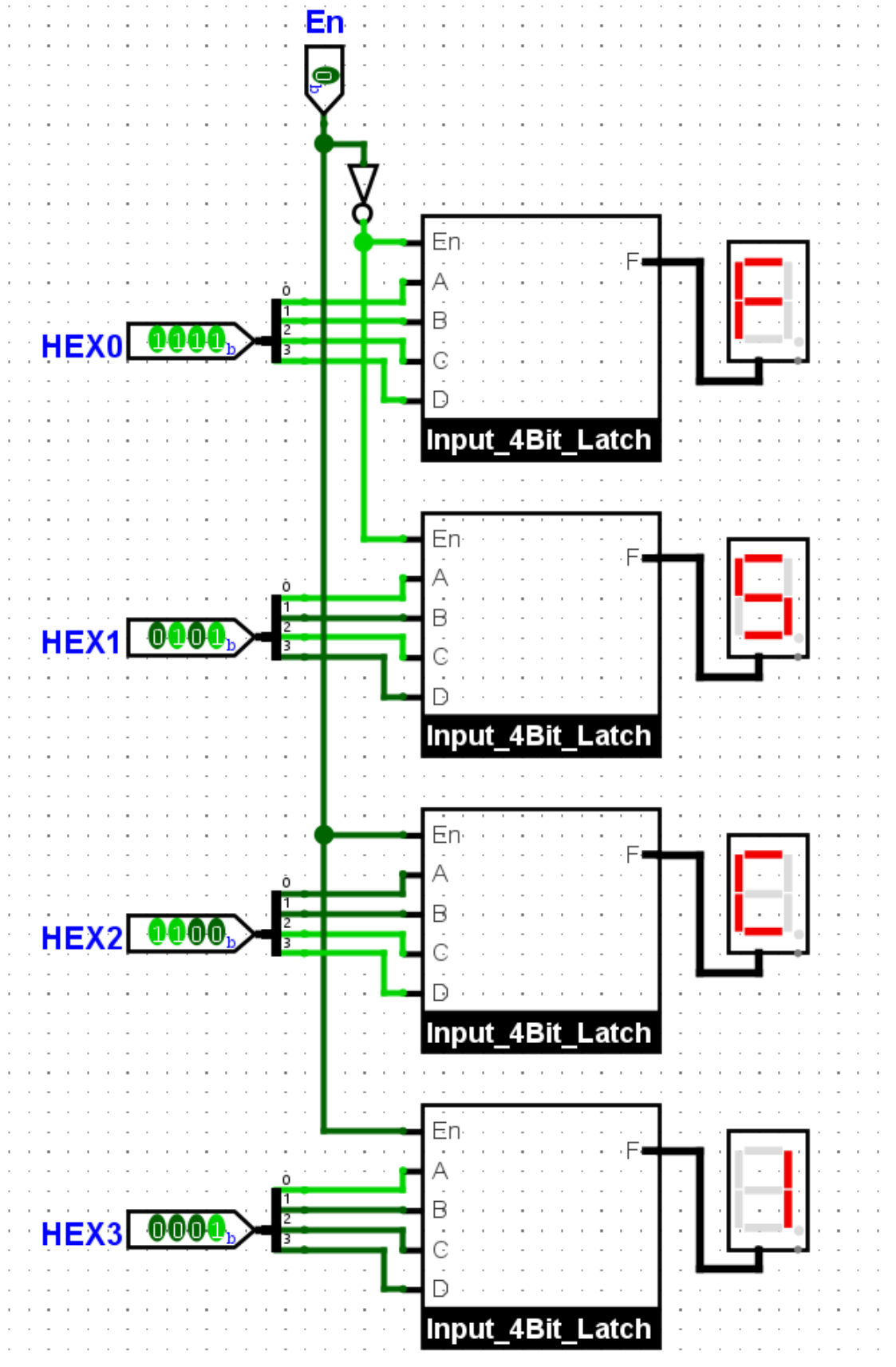


Figure 19 - Logisim controlling 4 HEX displays. Note: I am aware I used HEX displays and not 7segment here. I also should only have two four-bit inputs to mirror the actual solution.

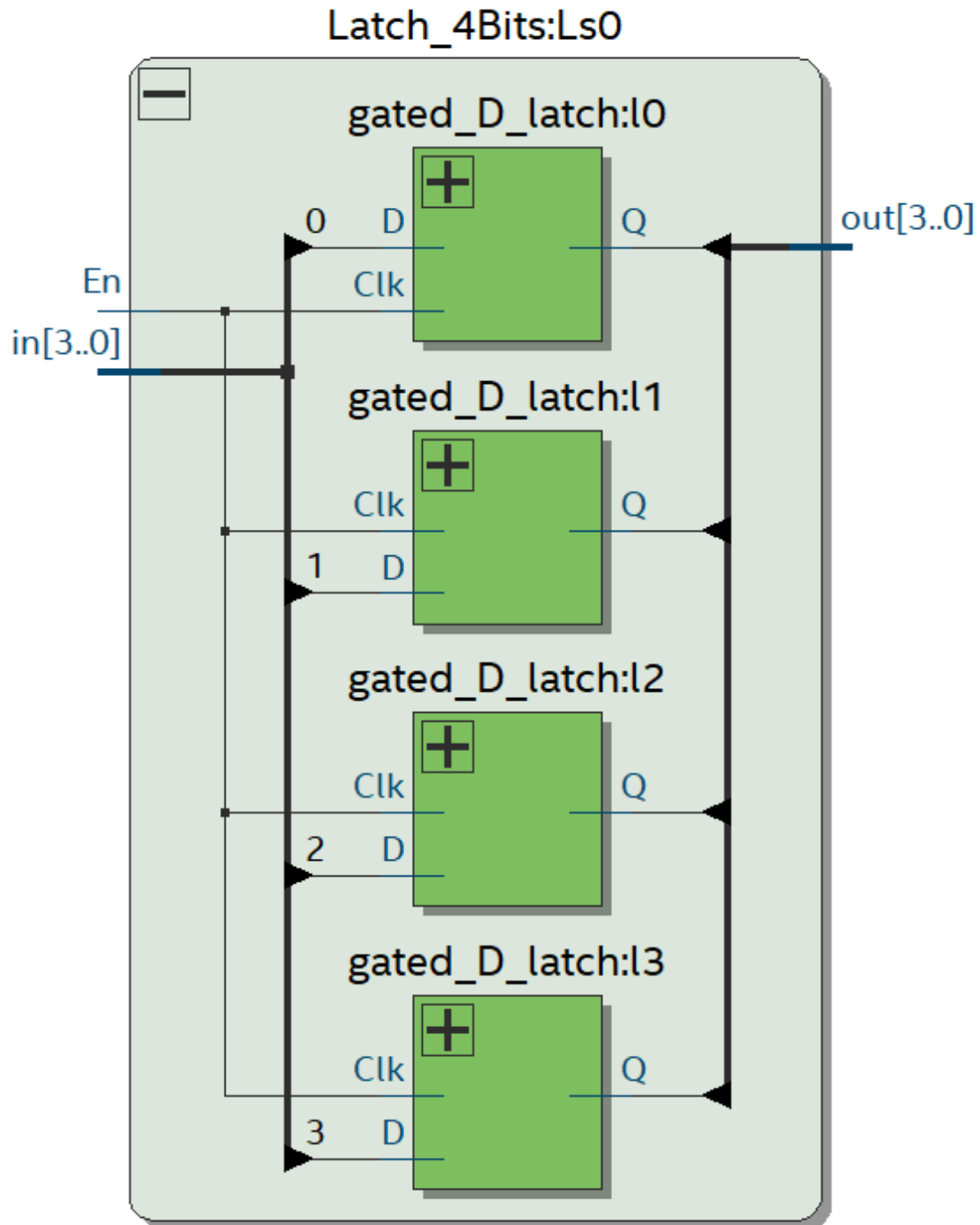


Figure 20 - Quartus 4-Bit D Latch RTL view

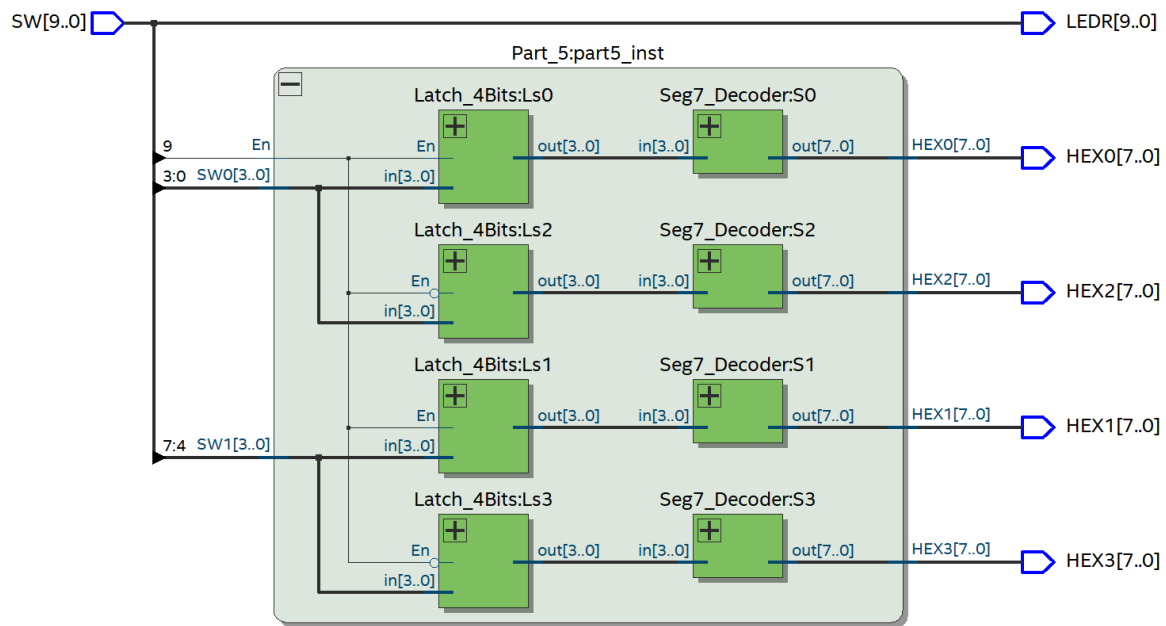


Figure 21 - Quartus RTL view of solution to control 4 segment displays using only 9 switches.

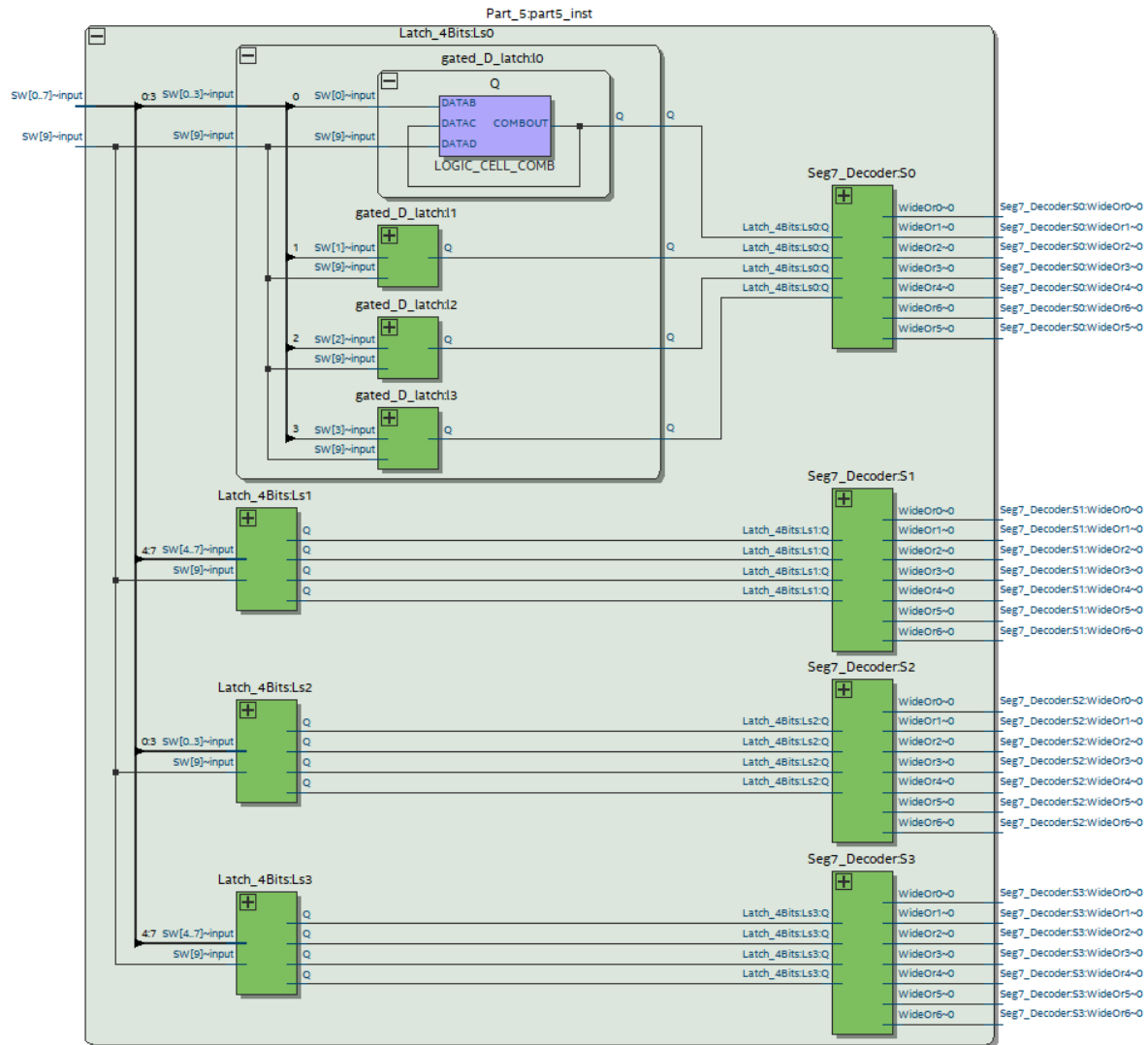


Figure 22 - Quartus Technology view of solution.

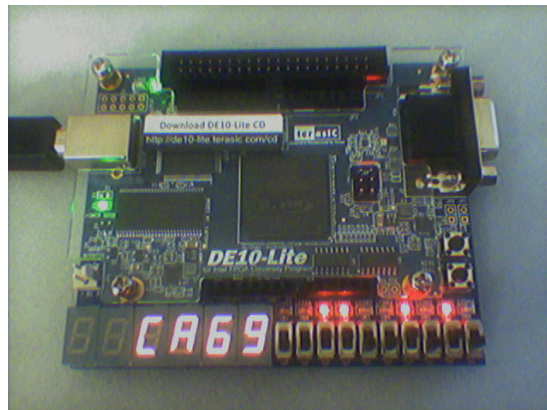


Figure 23 - Implementation showing Enable signal (SW9) low and input going to left two displays.

